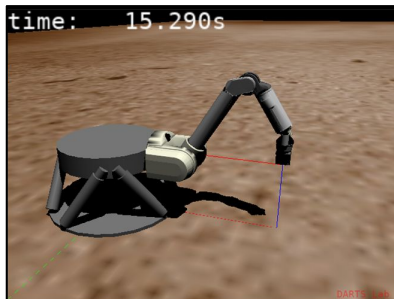


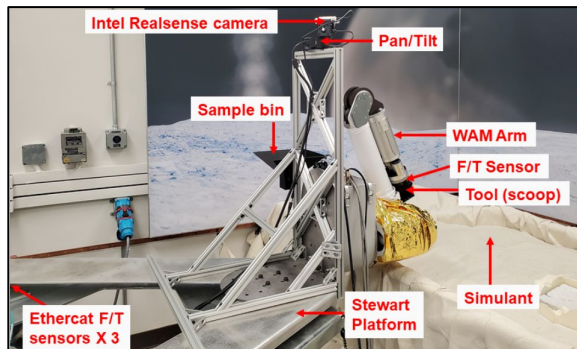
Erik Kramer | Ocean World Lander Autonomy Testbed

Project → A lander and robot arm sampling testbed to evaluate the performance of user autonomy algorithms

Tasks → Non-earth gravity **dynamics** emulation, **motion planning** algorithms, development of user features/**sequences**



Software Simulation and Hardware Operation



[Click](#) for Low-Gravity Emulation Video Demo

This research was carried out at the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration (80NM0018D0004)

Physical Emulation of Low-Gravity Ocean World Lander-Manipulator Dynamics on a Robotic Testbed

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Erica Tevere, and Hari Nayar



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California Institute of Technology

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Objectives

- Develop a mode to emulate non-earth gravity dynamics through torque offloading with software
- Solve kinematics issues causing Cartesian motion planner to find bad trajectories that result in faults
- Add user features and sequences as needed by autonomy teams

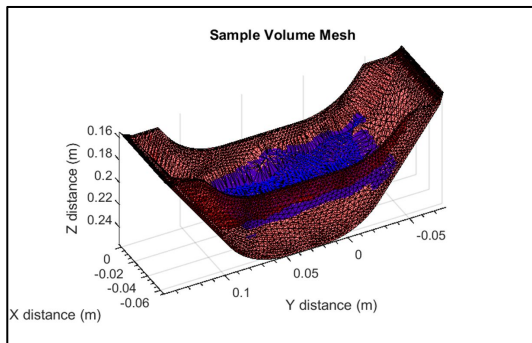
Process

- Analyzed telemetry with MATLAB to find root cause of robot faults
- Wrote new and updated old C++ code in a multi-author repository
- Utilized MATLAB, in house simulator, and hardware testing to verify methods

Results

- Demonstrated non-earth gravity dynamics through torque control
- Implemented a new motion planning optimizer that finds smooth trajectories and is 30% faster
- Added sequences such as radial scooping and features such a ROS interface for an autonomy subsystem

Testbed Feature Development



[Click](#) for Scoop Sequencing Video

